

Machine Learning based DDOS Detection

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Abstract- One of a high relentless attack is the crucial distributed DoS attacks. The types and tools for this attacks increases day-to-day as per the technology increases. So the methodology for detection of DDoS should be advanced. For this purpose we created an automated DDoS detector using ML which can run on any commodity hardware. The results are 98.5 % accurate. We use three classification algorithms KNN, RF and NB to classify DDoS packets from normal packets using two features, delta time and packet size. This detector mostly can detect all types of DDoS such as ICMP flood, TCP flood, UDP flood etc. In the older systems they detect only some types of DDoS attacks and some systems may require a large number of features to detect DDoS. Some systems may work only with certain protocols only. But our proposed model overcome these drawbacks by detecting the DDoS of any type without a need of specific protocol that uses less amount of features.

Index terms- Machine learning, DDoS detection

I. INTRODUCTION

Nowadays, the internet is everywhere. The systems are connected with one another by using various kinds of networks. A network can be defined as the n no.of computers, routers servers etc. A network can have both legal & illegal users (i.e) Hackers. Hacker is the one who exploits and access another one's data in illegal way. For this, a hacker can use various strategies. Those strategies are classified into two types, they are Active and Passive attacks. Passive attacks does not involve in modifying the resources, the hackers just sit back and analyze all the data. However in active attacks, hackers involve in the modification of the data as well as blocking a user from performing an action. One of the most famous and highly critical active attack is Denial of Service (Dos) attack. Dos attack is an active attack which crashes servers and systems on the network by flooding packets or requests in the network. Since the size of network is large today, there may be a very large no.of users available in the network. Therefore it becomes very hard to identify who are legal users and who are hackers. And also as the technology grows, the techniques for creating Dos attack also becomes increased. Identifying the Dos attack becomes more complex issue since there are various types of Dos attack strategies exist. Some of the types of the DoS attacks are ICMP flood, SYN flood, IP packet flood etc., To solve

this problem a machine learning based detector is a good solution.

II. LITERATURE REVIEW

Details about various types of distributed DoS are taken from . The network is analyzed and get packet's information from netflow protocol is shown in[1]. Netflow protocol is easy to get the packet information with less amount of effort. . But netflow protocol exist only in Cisco devices such as cisco routers. So it can't be used for other devices. The idea of using ML for distributed DoS detection is extracted from the paper[8]. From[11] it is useful to discover the various types of algorithms and it variations. IDS was developed with different machine learning algorithms, they are KNN, SVM, MLP and DT in [2]. Since MLP is one of the neural network based algorithm, it requires a large amount of data to train. It also very complex algorithm, so it takes long period of time to process. Decision Tree does not handle data with noise. But KNN deals with less amount of data with accuracy even with noisy data. So we consider KNN as one of the algorithm for this proposed model. Uygur Dinçalp1 [3] uses the rate of arriving various types of packet as the core concept for DDoS identification and also uses DBSCAN algorithm for clustering the data which is inefficient over live data capturing. So in this paper we consider the methodology of selecting features for ML model. Since only one algorithm is not enough for giving the best accurate results, we have to choose an additional classifier algorithms.[10]Shows the performance of the various classification algorithms which gives an idea to choose algorithm. [4] Shows that the accuracy of the Random Forest is greater than KNN, but slower than KNN. Therefore we choose both algorithms for this project. As in [5],[6] and [9] the precision of the naive bayes algorithm is more than both Random Forest and KNN.[7][15] Shows the accuracy of the Naive Bayesian. Therefore we add an additional algorithm that is, Naive bayes algorithm for classification.

III. CLASSIFICATION METHODS

A. Random Forest (RF)

The 'RF' is one of the most popular ML based classification model. It consist of large no. of decision trees(DT). The DT can make decision based on certain parameters. It have two types of nodes, parent and child

nodes. The topmost node is the root-node. The node at the ends are leaf-nodes. The leaf-nodes consist of decisions based on the conditions. The DT is highly sensitive to change in data. If any change done on the data without altering the conditions then the DT may lead to wrong decision. This problem was overcome in RF model. Since the RF have many DTs, each tree can work with independence on making decisions with random datasets. The random datasets is produced by modifying the occurrences of data without changing the length of the data. Therefore, all the trees receive equally sized data[8]. The correlation between the trees should be less to increase the accuracy of the output. If the correlation between the trees is high, then the propagation of wrongly made decision is also high, which reduces the accuracy of the algorithm. Then based on voting majority of all trees, the final decision is made.

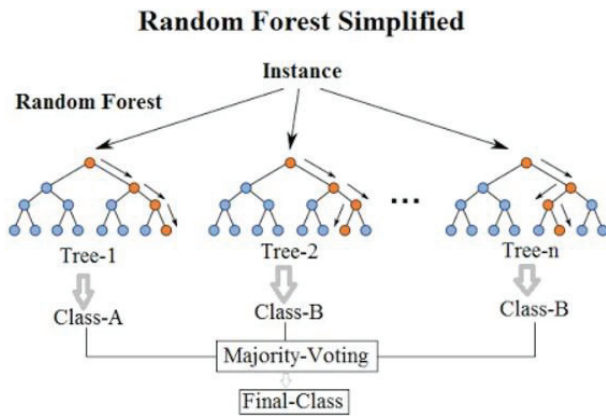


Fig. 1. Random Forest Information

B. KNN

The KNN method is a ML methods where this method works based on the nearest neighbors. This method is on the basis that similar objects are placed near each other. For this method to work efficiently the K has to be chosen carefully. When the K value is decreased, the stability of the model increases. When the K-value is increased, we can see the stability of this model increasing, but when the K-value crosses a certain limit and it has a value too high, the amount of errors also increases. So, it is best to choose the correct K-value for our desired problems. It can be chosen by running this model with various K-values and choose the best one with better stability and less errors. After choosing the K-value, we need to load the data and initiate the method. For each set of data, we need to satisfy the given query to acquire a result. After acquiring the result data set, we can choose the mean of the data as our best output.

C. Naive Bayesian

The Naïve Bayesian method is a probabilistic machine learning method that is used for the task of classification. It is based on the Bayes Theorem which is shown below[9]. Using the Bayes Theorem, we can identify the probability of one event happening such that a similar event has already occurred.

$$P(Y|X) = \frac{P(X|Y)P(Y)}{P(X)} \rightarrow \text{Posterior} = \frac{\text{likelihood} \times \text{prior}}{\text{evidence}}$$

$X = (X_1, X_2, \dots, X_k)$ represent k features. Y is the label with K possible values (class)

From a probabilistic perspective, $X_i \in X$ and Y are random variables

The value of X_i is x , Y is y .

This is a rather simple equation; we can't directly get the value of $P(Y|X)$ but we can get the value of $P(X|Y)$ and $P(Y)$ from the training data. The above part only consists of the Bayes theorem and it might be theoretically easy but in reality, it becomes much harder. So, we make a naïve assumption that all the features are independent. From this naïve assumption we can classify the parameters with fewer features. Since we have reduced the number of features, this method can handle any high-dimensional data really well.

IV. SOME TYPES OF DoS ATTACKS

A. SYN-Flood

Synchronous flood is one of the type of DoS attack, that breaks the 3-Way handshaking of TCP protocol during establishing communication between 2 parties. The client sends SYN_REQ (request) message to the server. The server responds with SYN_ACK (acknowledgement) to the client. Then the client sends ACK message to the server. This process is termed as 3-Way handshaking. When a hacker sends many SYN packets to the server at minimum intervals of time, the server sends SYN_ACK to the hacker. But the hacker does not send ACK to the client, leaving the connection halfly opened. Therefore for normal users, they don't have enough resources to allocate by the server. Therefore it causes the users to wait for an indefinite period of time[10].

B. ICMP-Flood

Normally, the client sends ICMP request to the server, the server sends ICMP reply to the client. For each ICMP request, the server creates memory allocation. If the hacker sends ICMP requests as flood to the server, the server suffers from overflow of memory which crashes the server.

V. PROPOSED SYSTEM

Steps to be followed

- Gather data using wireshark.
- Feature extraction.
- Data modification
- Classification
- Decision making

A. Tools

The wireshark is a tool which analyze the network and gives information about the packets such as packet length, time-stamp, source and destination etc, with the help of an API called Tshark. The required data can be collected by using pyshark, an interface for Tshark which allows the python program to communicate with the wireshark directly[11]. The manual DDoS attacks are made by us and the data-set are collected. We get data-set of previously happened real-time attacks from online data sources.

B. Data Preprocessing

Secondly, the data preprocessing should be done on the data to avoid noises and also helps to improve the performance of the said algorithm. The data preprocessing is done in two phases, they are feature extraction and modification of data to numerical values.

C. Feature Extraction

The feature extraction is the main thing for classification of data. In this project we consider three features, packet length, delta time and protocol.

D. Data Modification And Classification

Then these features are converted into numerical representation. The numerical values are given as input to the ML models, with 30% for training data and 70% for testing the model. Each ML model is processed simultaneously.

E. Decision Making

The decision is given by the voting majority of all the algorithms. Therefore the accuracy of this model will be great. We have used 3 different methods and since they produce 3 different nodes of result[12]. From these 3 result node, by voting, the best result is chosen.

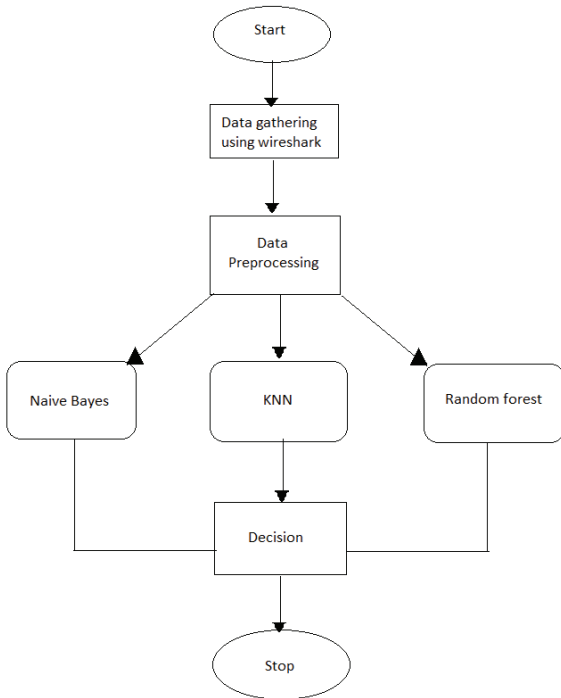


Fig. 2. Proposed System Architecture

F. Implementation

Getting the data set consist of source ip address, destination IP address , packet size and timestamp through pyshark, which is a python interface for wireshark .

G. Raw Data

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[Protocol:] TCP [Source IP:] 248.4.98.5 [Destination IP:]192.168.78.132 [Length:] 60 [Time :]1576337950.451037000
[Protocol:] TCP [Source IP:] 31.247.253.71 [Destination IP:]192.168.78.132 [Length:] 60 [Time :]1576337950.465745000
[Protocol:] TCP [Source IP:] 173.130.139.224 [Destination IP:]192.168.78.132 [Length:] 60 [Time :]1576337950.482473000
[Protocol:] TCP [Source IP:] 52.241.254.120 [Destination IP:]192.168.78.132 [Length:] 60 [Time :]1576337950.496335000
[Protocol:] TCP [Source IP:] 211.129.200.110 [Destination IP:]192.168.78.132 [Length:] 60 [Time :]1576337950.510097000
[Protocol:] TCP [Source IP:] 108.10.33.92 [Destination IP:]192.168.78.132 [Length:] 60 [Time :]1576337950.523913000
[Protocol:] TCP [Source IP:] 95.185.76.107 [Destination IP:]192.168.78.132 [Length:] 60 [Time :]1576337950.537977000
[Protocol:] TCP [Source IP:] 176.62.242.72 [Destination IP:]192.168.78.132 [Length:] 60 [Time :]1576337950.552557000
[Protocol:] TCP [Source IP:] 164.35.60.195 [Destination IP:]192.168.78.132 [Length:] 60 [Time :]1576337950.566032000
  
```

Fig. 3. Raw Data Information

Then the feature extraction was made. The considered features are Delta value, packet size and DDoS label[13].

Delta value is calculated by finding the difference of timestamps between two consecutive packets flowed in a network.

$$\Delta t = t_2 - t_1$$

Where,

t_1 =timestamp of first arriving packet

t_2 =timestamp of secondly arriving packet

The data is converted into numerical format by using transform function, which is a implementation from scikit-learn.

Then the encoded data is given as a test_data_split function to split data into training and testing set separately[14]. Then each model is trained using the training data and tested by using the test data. The formulae for 3 algorithms are given below

Naive Bayes

$$P(y|X) = P(X|y) * P(y) / P(X)$$

KNN

Distance calculation

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

```

[Protocol:] TCP [Source IP:] 187.11.159.209 [Destination IP:]192.168.78.132 [Length:] 60 [Time :]1576337949.013975000 [Label :]Yes
[Protocol:] TCP [Source IP:] 17.227.230.10 [Destination IP:]192.168.78.132 [Length:] 60 [Time :]1576337949.026457000 [Label :]Yes
[Protocol:] TCP [Source IP:] 187.156.125.11 [Destination IP:]192.168.78.132 [Length:] 60 [Time :]1576337949.039509000 [Label :]Yes
[Protocol:] TCP [Source IP:] 146.34.254.120 [Destination IP:]192.168.78.132 [Length:] 60 [Time :]1576337949.050748000 [Label :]Yes
[Protocol:] TCP [Source IP:] 202.47.171.141 [Destination IP:]192.168.78.132 [Length:] 60 [Time :]1576337949.061258000 [Label :]Yes
[Protocol:] TCP [Source IP:] 58.107.186.206 [Destination IP:]192.168.78.132 [Length:] 60 [Time :]1576337949.074846000 [Label :]Yes
[Protocol:] TCP [Source IP:] 226.50.187.12 [Destination IP:]192.168.78.132 [Length:] 60 [Time :]1576337949.089728000 [Label :]Yes
[Protocol:] TCP [Source IP:] 149.151.6.84 [Destination IP:]192.168.78.132 [Length:] 60 [Time :]1576337949.102827000 [Label :]Yes
[Protocol:] TCP [Source IP:] 115.192.27.111 [Destination IP:]192.168.78.132 [Length:] 60 [Time :]1576337949.115811000 [Label :]Yes
[Protocol:] TCP [Source IP:] 11.56.3.224 [Destination IP:]192.168.78.132 [Length:] 60 [Time :]1576337949.127969000 [Label :]Yes
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[Protocol:] TCP [Source IP:] 237.205.182.205 [Destination IP:]192.168.78.132 [Length:] 60 [Time :]1576337949.163046000 [Label :]Yes
[Protocol:] TCP [Source IP:] 211.213.162.109 [Destination IP:]192.168.78.132 [Length:] 60 [Time :]1576337949.175201000 [Label :]Yes
[Protocol:] TCP [Source IP:] 145.135.0.3 [Destination IP:]192.168.78.132 [Length:] 60 [Time :]1576337949.186277000 [Label :]Yes
[Protocol:] TCP [Source IP:] 190.16.212.92 [Destination IP:]192.168.78.132 [Length:] 60 [Time :]1576337949.196690000 [Label :]Yes
[Protocol:] TCP [Source IP:] 44.13.122.119 [Destination IP:]192.168.78.132 [Length:] 60 [Time :]1576337949.208527000 [Label :]Yes
[Protocol:] TCP [Source IP:] 13.229.91.143 [Destination IP:]192.168.78.132 [Length:] 60 [Time :]1576337949.221381000 [Label :]Yes
[Protocol:] TCP [Source IP:] 135.135.62.62 [Destination IP:]192.168.78.132 [Length:] 60 [Time :]1576337949.233221000 [Label :]Yes
[Protocol:] TCP [Source IP:] 236.126.19.214 [Destination IP:]192.168.78.132 [Length:] 60 [Time :]1576337949.243508000 [Label :]Yes
[Protocol:] TCP [Source IP:] 170.154.146.45 [Destination IP:]192.168.78.132 [Length:] 60 [Time :]1576337949.255709000 [Label :]Yes
[Protocol:] TCP [Source IP:] 110.210.20.222 [Destination IP:]192.168.78.132 [Length:] 60 [Time :]1576337949.268397000 [Label :]Yes
[Protocol:] TCP [Source IP:] 160.68.227.73 [Destination IP:]192.168.78.132 [Length:] 60 [Time :]1576337949.279295000 [Label :]Yes
[Protocol:] TCP [Source IP:] 110.218.30.238 [Destination IP:]192.168.78.132 [Length:] 60 [Time :]1576337949.292994000 [Label :]Yes
[Protocol:] TCP [Source IP:] 197.232.135.254 [Destination IP:]192.168.78.132 [Length:] 60 [Time :]1576337949.306196000 [Label :]Yes
  
```

Fig. 4. Training data Information

TABLE I. ACCURACY TABLE

Machine Learning	Accuracy
Naive Bayesian	97.65%
K-means clustering	99.88%
Random Forest	100%

Then the decision is taken by majority voting method for all the algorithms.

VI. CONCLUSION

By using the traditional techniques, most of the work needs to be done manually and also the accuracy of the result is low and the identification of the risk also takes a long time. But by using a Machine Learning way of approach, we can identify the threats easily and way faster than the traditional methods. In our proposal we have used three ML algorithms Naive Bayesian, KNN and Random Forest. The Reason for

choosing these three algorithms is that these three algorithms require less attributes and a low amount of training data to run the detection process compared to other machine learning algorithms. What we do is we choose the best two result from the three outputs for detection which increases the accuracy of the process and also it needs only a small amount of resource power. Since it needs less resource power, it can run even on a commodity hardware. Thus, this method can detect DDoS attacks of any type with less amount of time and more accuracy.

VII. SCOPE AND LIMITATIONS

Since our detection model has trained with some of the popular DDoS tools such as hping3, it may not detect DDoS attacks which are created by other DDoS tools. But it can be trained easily for any type of DDoS attacks, so its detecting capacity can be extended. Therefore no need of creating a separate detector to find other DDoS attacks. It does not need a large data to detect the DDoS attack.

VIII. FUTURE ENHANCEMENT

Since it is a low resource consumption and it can run on a low commodity hardware, we will deploy it on any network or security devices such as routers, firewalls or on a server. This model only detects the DDoS attacks, but in future we will extend this to a DDoS avoiding model, which enhances the security of the server. We will increase the efficiency of this model as per the technology growth.

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